HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and developed scalable Python data pipelines for batch processing, ensuring data integrity and optimizing performance for large datasets.
- Engineered and maintained production-grade Python codebases, focusing on reliability and maintainability within a substantial enterprise environment.
- Implemented batch processing workflows that processed and transformed large datasets, meeting stringent performance and reliability requirements.
- Collaborated with senior software engineers and technical stakeholders to define requirements and deliver high-quality data processing solutions.
- Troubleshot and optimized production applications, improving system efficiency and resolving complex technical challenges.
- Contributed to the design and implementation of data infrastructure, enhancing scalability and maintainability of existing systems.
- Deployed containerized Python applications to cloud-based infrastructure, managing configurations and ensuring smooth integration with distributed systems.

SKILLS

Programming Languages – Python, Java, SQL, C#

Data Engineering Tools – Apache Spark, PySpark, Databricks, Apache Airflow, Delta Lake, Apache Hudi

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch), Google Cloud Platform, Azure

Databases – PostgreSQL, MySQL, Oracle, Relational Databases

Software Development – Data Structures, Algorithms, Testing, System Design, CI/CD, Git, Jenkins

Concepts – Distributed Data Processing, Large-scale ETL/ELT, Batch Processing, Data Integrity, Performance Optimization

Observability – Splunk, Dynatrace, CloudWatch

PROJECTS

Enterprise Data Pipeline Refactoring (HCL Technologies)

- Re-architected and optimized existing Python ETL pipelines for a financial services client, reducing data processing times by an average of **30%**.
- Implemented robust error handling and logging mechanisms within the data pipelines, improving system observability and simplifying incident response.
- Migrated data processing logic to leverage **AWS Lambda** and **S3** for more scalable and cost-effective batch operations.

Distributed Systems Data Processing Framework (MSc Project)

- Developed a Python framework for distributed data processing, simulating large-scale ETL/ELT operations using concepts from **Apache Spark**.
- Designed and implemented data validation routines to ensure data integrity across distributed datasets, catching approximately **85%** of common data quality issues.
- Built interactive dashboards using **Power BI** to visualize processing metrics and data quality reports for stakeholders.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)